Name: _____

WORKSHEET

Tree diagrams

1 An unbiased coin is flipped and a fair six-sided die is rolled.
Draw a tree diagram showing all possible outcomes and use it to find:

a $P(\text{H and } 5)$

b $P(\text{T and an even number})$

c $P(\text{an odd number})$

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2 A three course meal offers the choice of 2 entrees, 3 mains and 2 desserts.
Draw a tree diagram showing all possible outcomes.

a How many outcomes are there?

b If you randomly select each course, what is the probability of choosing any given three course meal?

3 Draw a tree diagram showing the outcomes for flipping three unbiased coins.
Use your tree diagram to answer the following:

a $P(\text{TTH})$

b $P(\text{exactly 1 Head})$

c $P(\text{more than 1 Head})$

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d P(less than 3 Tails)

4 A fair four-sided die is rolled twice. Draw a tree diagram and use it to find:

a P(1 then 1)

b P(3 then 4)

c P(a 3 and a 4 in any order)

d P(the same number)

e P(not the same number)

f P(exactly 1 odd number)

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5 A circular spinner has 3 parts red, 4 parts blue and 5 parts white.

a Draw a tree diagram for spinning two times.

b What is the probability of spinning white and white?

c What is the probability of spinning red and red?

c What is the probability of spinning white and blue in that order?

d What is the probability of spinning blue exactly once?

6 A bag contains three $10 banknotes, three $20 banknotes, and two $50 banknotes.

a Draw a tree diagram showing probabilities for randomly selecting one banknote.

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b Draw a tree diagram for selecting two banknotes, where the first banknote is returned to the bag before drawing the second banknote. Use your tree diagram to find:

i $P(\$50, \$50)$

ii $P(\$20, \$10$ in that order)

iii P(ending up with \$70)

iv P(the two notes are the same)

c Draw a tree diagram for selecting two banknotes, where the first banknote is not replaced before drawing the second banknote. Find:

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i $P(\$50, \$50)$

ii $P(\$20, \$10$ in that order$)$

iii $P($ending up with \$70$)$

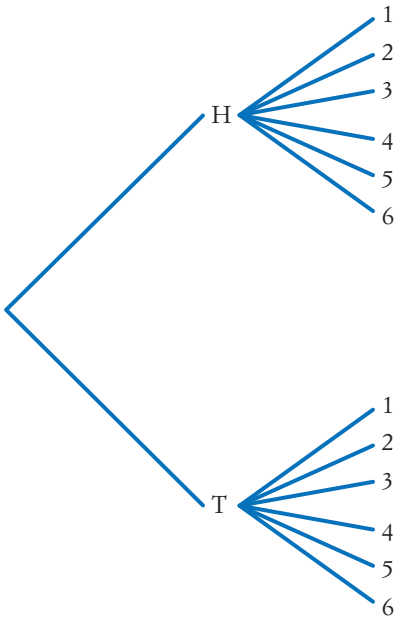
iv $P($the two notes are the same$)$

Answers

1 **a** $\frac{1}{12}$

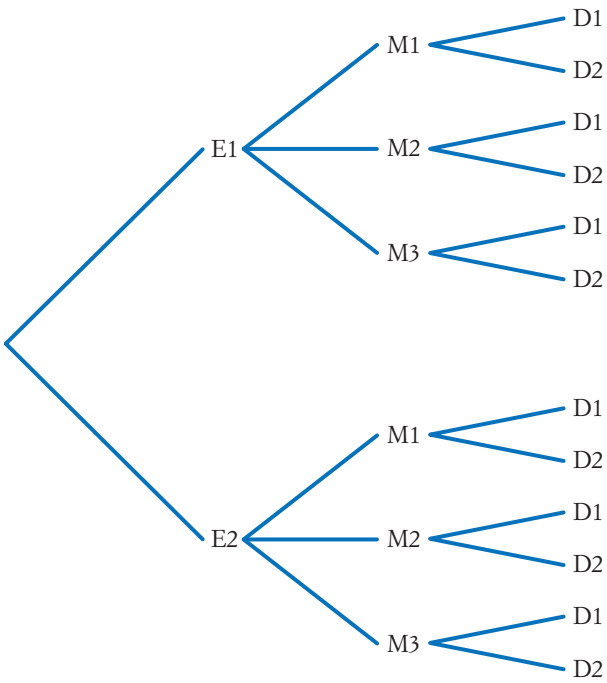
b $\frac{1}{4}$

c $\frac{1}{2}$



2 **a** 12

b $\frac{1}{12}$



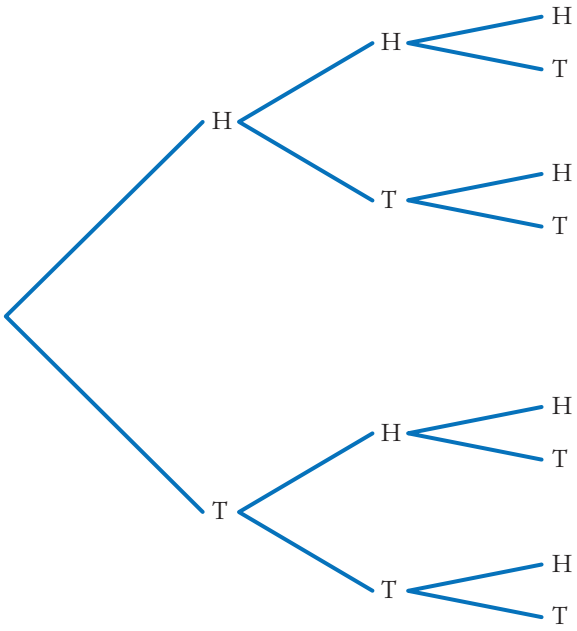
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3 **a** $\frac{1}{8}$

b $\frac{3}{8}$

c $\frac{1}{2}$

d $\frac{7}{8}$



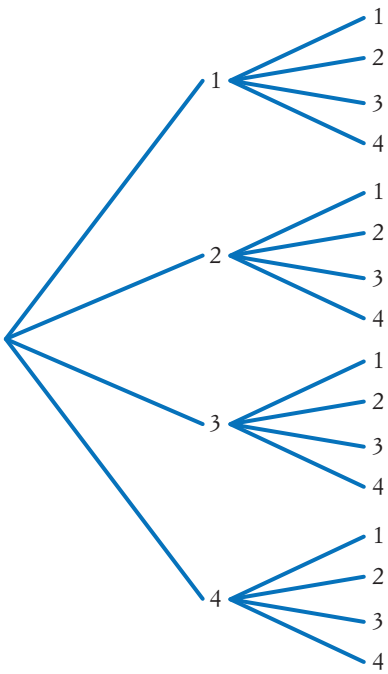
4 **a** $\frac{1}{16}$

b $\frac{1}{16}$

c $\frac{1}{8}$

d $\frac{1}{4}$

e $\frac{3}{4}$



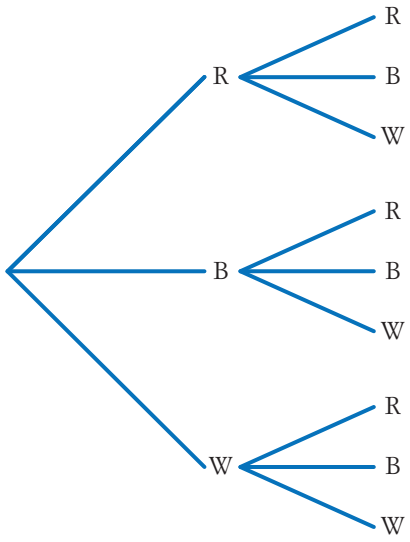
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5 a $\frac{25}{144}$

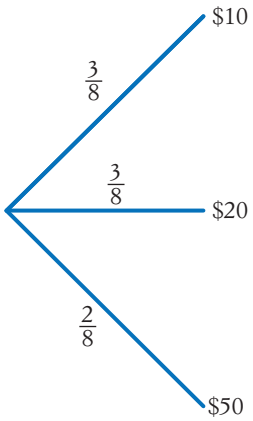
b $\frac{9}{144}$

c $\frac{5}{36}$

d $\frac{4}{9}$



6 a

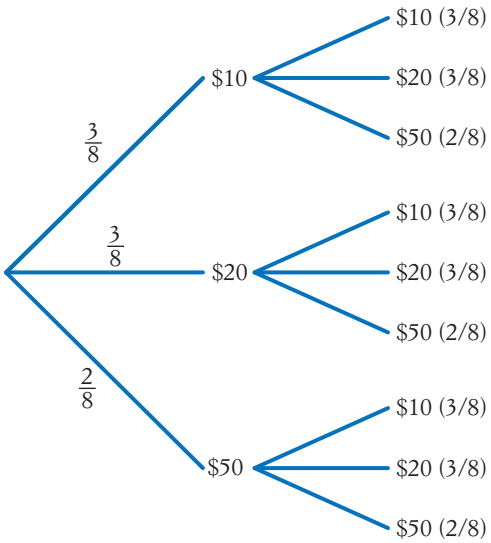


b i $\frac{1}{16}$

ii $\frac{9}{64}$

iii $\frac{3}{16}$

iv $\frac{11}{32}$



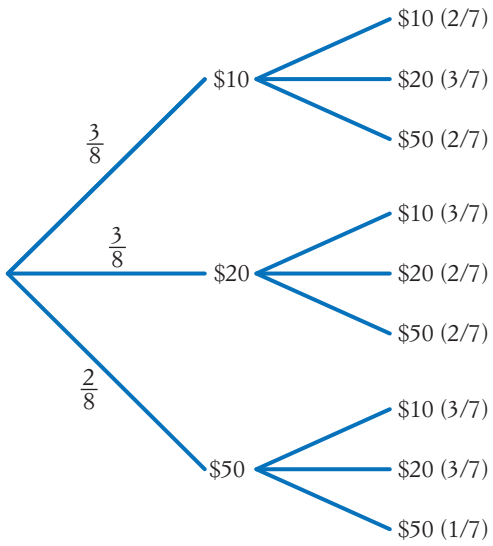
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c **i** $\frac{1}{28}$

ii $\frac{9}{56}$

iii $\frac{3}{14}$

iv $\frac{1}{4}$



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